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Paper Name: Network-based discovery of bioactive polyphenols based on transcriptomic changes across the colorectal adenoma–carcinoma sequence

Network-based discovery of bioactive polyphenols based on transcriptomic changes across the colorectal adenoma–carcinoma sequence

Abstract

Colorectal adenomas are major precancerous lesion of colorectal cancer (CRC) and provide a clinically actionable window for preventing invasive CRC. However, the molecular changes associated with adenoma progression remain incompletely defined. Therefore, elucidating the mechanisms underlying adenoma carcinogenesis to identify druggable targets is an important strategy for CRC control. We analyzed 33 colorectal tissue samples, comprising adenomas from 11 individuals and paired stage-I CRC and adjacent normal tissues from an additional 11 patients. Using a network-based integrative approach, we identified stage-associated genes underlying the mechanism of adenoma carcinogenesis. We identified 725 differentially expressed genes (DEGs) in adenoma versus normal tissue, 312 in CRC versus adenoma, and 1,544 in CRC versus normal tissue. Twenty-nine genes changed concordantly across the two sequential comparisons; all 29 mapped to the interactome used for proximity analysis. Adenoma formation was associated with inflammatory recruitment and loss of differentiated epithelial and metabolic functions, whereas CRC showed additional extracellular matrix, focal adhesion, and cell-cycle activity. Of the 66 compound entries screened, 11 unique compounds are identified. Kaempferol showed the strongest network proximity, followed by 4-methylcatechol, quercetin, punicalagin, and umbelliferone. Collectively, this study provides a novel network medicine-based approach applicable to the prevention and treatment of CRC.

Keywords: colorectal adenoma; colorectal cancer; transcriptomics; network medicine; polyphenols; molecular docking

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1. Introduction

Colorectal cancer (CRC) remains a major global health burden. GLOBOCAN 2022 estimated more than 1.9 million new CRC cases and approximately 904,000 deaths worldwide, making CRC the third most frequently diagnosed cancer and the second leading cause of cancer death (Bray et al., 2024). In China, an estimated 517,100 new CRC cases and 240,000 deaths occurred in 2022 (Han et al., 2024).

Most sporadic CRCs arise through multistage transformation of non-neoplastic colorectal epithelium into an adenoma and invasive carcinoma. The conventional adenoma–carcinoma pathway accounts for approximately 90% of the sporadic CRC precursors and is commonly initiated by APC/WNT dysregulation and followed by genetic and epigenetic alterations affecting RAS–MAPK, PI3K, TGF-β and TP53-associated processes (Keum and Giovannucci, 2019). Importantly, progression from a premalignant lesion to invasive cancer generally requires at least 10 years. Thus, there is a unique window of opportunity for secondary prevention of CRC (Keum and Giovannucci, 2019; Nguyen et al., 2020).

Current prevention takes advantage of this window mainly through colonoscopic detection and endoscopic removal of adenomas (Kaltenbach et al., 2020), yet with limited options for directly interfering with the molecular processes that drive adenoma progression. After removing visible adenomas, new metachronous adenomas may still develop. A meta-analysis of placebo groups in post-polypectomy trials estimated that approximately 60% of patients had adenoma recurrence or detection within five years, and about 14% developed advanced adenomas during the same period (Shi et al., 2017). These findings suggest that endoscopic removal alone may be insufficient and that additional preventive strategies are needed.

Polyphenols are a diverse group of plant-derived natural compounds that have attracted increasing attention as potential agents for colorectal cancer prevention and treatment, such as polyphenol-based nanoparticles (Wahnou et al., 2023). Experimental studies have shown that polyphenols exhibit multiple bioactive properties that affect processes closely related to tumor development, including inflammation, oxidative stress, cell proliferation, apoptosis, and cancer-associated signaling pathways, which matches the biological complexity of adenoma carcinoma progression (Wahnou et al., 2023).

In addition, network pharmacology offers a systematic approach for connecting candidate compounds, predicted targets, disease-related genes, protein–protein interaction networks, and enriched biological pathways (Zhou et al., 2020; Zhao et al., 2023). Applying network pharmacology to colorectal adenoma carcinoma progression may help identify critical pathway modules and potential polyphenol compounds involved in colorectal carcinogenesis, providing a basis for further experimental validation.

2. Materials and methods

2.1 Data source and study design

Processed RNA-sequencing expression data were obtained from GEO accession GSE228012 (SRA PRJNA940523) by Lin et al. (2025). The dataset comprised adenomas from 11 individuals and paired stage-I CRC and adjacent normal tissues from an additional 11 patients. Thus, the adenoma group was independent of the CRC–normal patient pairs. GEO is maintained as a public functional-genomics repository (Edgar et al., 2002).

The compound–target resource followed the curation workflow of do Valle et al. (2021). Food polyphenols were identified from Phenol-Explorer (Neveu et al., 2010),

mapped to PubChem, filtered using Comparative Toxicogenomics Database disease annotations (Davis et al., 2023), and required experimentally supported protein-binding information in STITCH 5 (Szklarczyk et al., 2016). The source workflow reported 65 polyphenols and 598 protein targets; the present proximity output contained 66 rows because one compound representation was duplicated before identity normalization.

2.2 Discovery of molecular mechanisms related to colorectal carcinogenesis

The processed FPKM expression matrix was downloaded from the GEO supplementary files and analyzed using R. Duplicate gene symbols were removed by retaining the gene with the highest average expression level. Genes with low expression levels were filtered out, and expression values were transformed using log2(FPKM+1) before downstream analysis.

Differential expression analysis was performed using the limma package (Ritchie et al., 2015). Three comparisons were conducted: CRC versus normal, adenoma versus normal, and CRC versus adenoma. Genes with a Benjamini–Hochberg-adjusted P value < 0.05 and $|\log2 \text{ fold change}| > 1$ were considered significantly differentially expressed. Principal component analysis (PCA) was performed using FactoMineR and visualized with factoextra (Lê et al., 2008; Kassambara and Mundt, 2020) based on the log2-transformed expression matrix. Volcano plots were generated using ggplot2 (Wickham, 2016) to show upregulated, downregulated and non-significant genes in each comparison.

The results were further screened for genes associated with adenoma–carcinoma progression. Genes significantly upregulated in both the normal-to-adenoma and adenoma-to-CRC transitions were defined as progressively upregulated, whereas genes significantly downregulated in both transitions were defined as progressively downregulated. This yielded 29 progression-associated genes (8 up–up and 21 down–down). Heatmaps were constructed using pheatmap (Kolde, 2019).

Functional enrichment analysis was performed using clusterProfiler (Yu et al., 2012). Gene symbols were converted to Entrez identifiers using the org.Hs.eg.db annotation package (Carlson, n.d.). Gene Ontology biological-process, cellular-component and molecular-function analyses used the mapped genes from each complete DEG set (Gene Ontology Consortium, 2021), whereas KEGG analysis was performed separately for mapped upregulated and downregulated genes (Kanehisa et al., 2021). Terms and pathways with Benjamini–Hochberg-adjusted $P < 0.05$ were considered significant.

2.3 PPI construction

The 29 progression-associated genes were submitted to STRING, with the organism restricted to Homo sapiens (Szklarczyk et al., 2023). The interaction network was imported into Cytoscape for visualization and topological analysis (Shannon et al., 2003). Degree, betweenness centrality and closeness centrality were calculated for each node; genes with the highest degree values were treated as candidate hubs. The functional-association network was set at confidential scores of 0.150.

The score-filtered STRING node and edge tables were imported into Cytoscape for visualization and topological analysis (Shannon et al., 2003). Degree was defined as the number of edges incident on a node; betweenness centrality as the proportion of shortest paths passing through a node; and closeness centrality as the reciprocal of mean shortest-path distance to reachable nodes. Centrality was calculated on the same score-filtered graph.

2.4 Network proximity between polyphenol targets and colorectal cancer protein

To connect the transcriptomic alterations identified across the adenoma–carcinoma sequence with candidate polyphenols, we applied the network-proximity framework described by Guney et al. (2016) and subsequently adapted for polyphenols by do Valle et al. (2021). The analysis was performed on the human interactome assembled by do Valle et al. (2021), which integrates experimentally supported physical protein–protein interactions from 16 databases and contains a largest connected component of 17,651 proteins connected by 351,393 interactions. In this network, proteins are represented as nodes and experimentally supported protein–protein interactions as edges.

The 29 progression-associated genes identified from the transcriptomic analysis were mapped to their corresponding proteins in this interactome. The experimentally supported targets of each polyphenol were mapped to the same network. Network proximity was then used to quantify how closely the polyphenol targets were positioned relative to the progression-associated proteins.

For each polyphenol target, the shortest path to the nearest progression-associated protein was calculated. The network proximity was defined as:

$$d_c(S,T) = \frac{1}{\mid T \mid}\sum_{t\in T} \quad min_{s\in S}\, d(s,t)$$

where $d(s,t)$ represents the shortest path length between a progression-associated protein $s$and a polyphenol target t. A smaller $d_c$value indicated that the targets of the polyphenol were located closer to the progression-associated protein network.

To determine whether the observed proximity was greater than expected by chance, 1,000 degree-preserving randomizations were performed. In each iteration, two random protein sets matching the sizes and degree distributions of the original 29 progression-associated proteins and polyphenol targets were generated, and their network proximity was recalculated. The mean $\mu_{d_c}$and standard deviation $\sigma_{d_c}$of the random distribution were used to calculate a normalized proximity score:

$$Z_{d_c} = \frac{d_c - \mu_{d_c}}{\sigma_{d_c}}$$

A negative Z_{d_c}value indicated that the polyphenol targets were closer to the progression-associated proteins than expected by chance. Polyphenols with smaller $d_c$values and more negative Z_{d_c}scores were considered more closely associated with the molecular network underlying adenoma–carcinoma progression.

2.5 Molecular docking evaluation

The crystal structures of colorectal cancer related proteins were obtained from the RCSB Protein Data Bank (Niederhut et al., 2001; Boyles et al., 2020). Protein structures were prepared using AutoDock Tools 1.5.6 (Morris et al., 2009) by removing water molecules and crystallographic ligands, adding polar hydrogen atoms, assigning Gasteiger charges, and exporting the receptors in PDBQT format. In the current analysis, NAD and zinc ions were not retained in the prepared 1HT0 structure, whereas the complete chain A–F assembly of 6WZM was used as the receptor.

The three-dimensional structures of the selected phenolic compounds were downloaded from PubChem in SDF format and converted using Open Babel (O'Boyle et al., 2011). The ligands were subsequently prepared in AutoDock Tools by assigning Gasteiger charges and atom types and defining eligible rotatable bonds as flexible

before conversion to PDBQT format. Molecular docking was performed using AutoDock Vina 1.1.2 (Trott and Olson, 2010). The docking search space was defined using a grid centre of original ligand. The exhaustiveness was set to 8, and the binding modes were generated for each ligand. For each receptor–ligand pair, the pose with the lowest predicted Vina score was selected for interaction analysis. Protein–ligand interactions were visualized using BIOVIA Discovery Studio Visualizer version. Docking scores were expressed in kcal/mol and used only for relative candidate prioritization within each receptor system.

3. Results

3.1 Transcriptomic changes during adenoma carcinoma progression

To capture the molecular features of colorectal adenoma-carcinoma progression, we analyzed transcriptomic data spanning the adenoma-carcinoma sequence. The dataset contained expression profiles for 18,116 genes across 33 colorectal tissue samples, including 11 normal samples, 11 adenoma samples, and 11 CRC samples. Principal component analysis was performed to assess the overall transcriptomic differences among normal colorectal tissue, adenoma, and CRC samples. Normal samples formed a relatively compact cluster and were clearly separated from most adenoma and CRC samples. Adenoma samples occupied an intermediate region, whereas CRC samples displayed a broader distribution and partially overlapped with the adenoma group. These results indicate that substantial transcriptomic changes had already occurred at the adenoma stage, followed by additional heterogeneous alterations compared to mature CRC samples (Fig. 1A).

Differential expression analysis revealed obvious transcriptional changes at different stages of colorectal tumorigenesis. Substantial transcriptional alterations were already evident in adenomas relative to normal tissue, and we identified in total 725 differentially expressed genes. Among the most prominently downregulated genes were CA1, CHGA, GUCA2A, SLC26A3, and ZG16, whereas MMP7, TCN1, DUOXA2, DEFA6, and REG4 showed increased expression (Fig. 1B). The comparison between CRC and adenoma identified 312 differentially expressed genes associated with malignant progression. THBS2, CXCL8, COL1A1, PCSK9, and EREG were among the prominently upregulated genes in CRC, while CLCA1, FCGBP, DEFA6, SPINK4, and DEFA5 were downregulated. The genes highlighted in this comparison differed partly from those identified during adenoma formation, suggesting that the two transitions involve largely different transcriptional changes (Fig. 1C). The direct comparison between CRC and normal tissue revealed the accumulated expression changes across the complete disease sequence. CDH3, CEMIP, CXCL8, MMP7, and DPEP1 were markedly upregulated in CRC, whereas CA1, AQP8, CHGA, ZG16, and SLC26A3 were among the most strongly downregulated genes. Several of these genes were also altered in the sequential comparisons, indicating that part of the transcriptional disturbance initiated at the adenoma stage was maintained or further enhanced in CRC (Fig. 1D).

However, we observed that not all differentially expressed genes followed a continuous pattern across the three stages. For example, DEFA6 was upregulated in adenoma relative to normal tissue but downregulated in CRC relative to adenoma. This stage specific change suggests that some genes may be associated primarily with adenoma formation solely, but not during the whole development. To identify genes more closely associated with continuous disease progression, genes showing concordant expression changes across normal tissue, adenoma, and CRC were further screened. A total of 29 genes exhibited progressive expression patterns, including eight

progressively upregulated genes and 21 progressively downregulated genes. The progressively upregulated genes were CXCL2, SLC22A3, HIST3H2A, SLCO4A1, MYC, COL12A1, TNFRSF12A, and CXCL8 (Fig. 2A), while down regulated genes included JCHAIN, ST6GALNAC6, PADI2, ADH1C, LYPD8, HSD17B2, MFSD4A, GPT, TNFRSF17, SCIN, BEST2, and ADTRP (Fig. 2B).

Together, these results revealed both stage-specific changes and a smaller set of 29 genes that change consistently across all three stages. Here, we use these 29 genes as the progression-associated set for the network analyses subsequently. To understand the functional changes behind these expression patterns, we next asked which biological processes and pathways are affected at each stage.

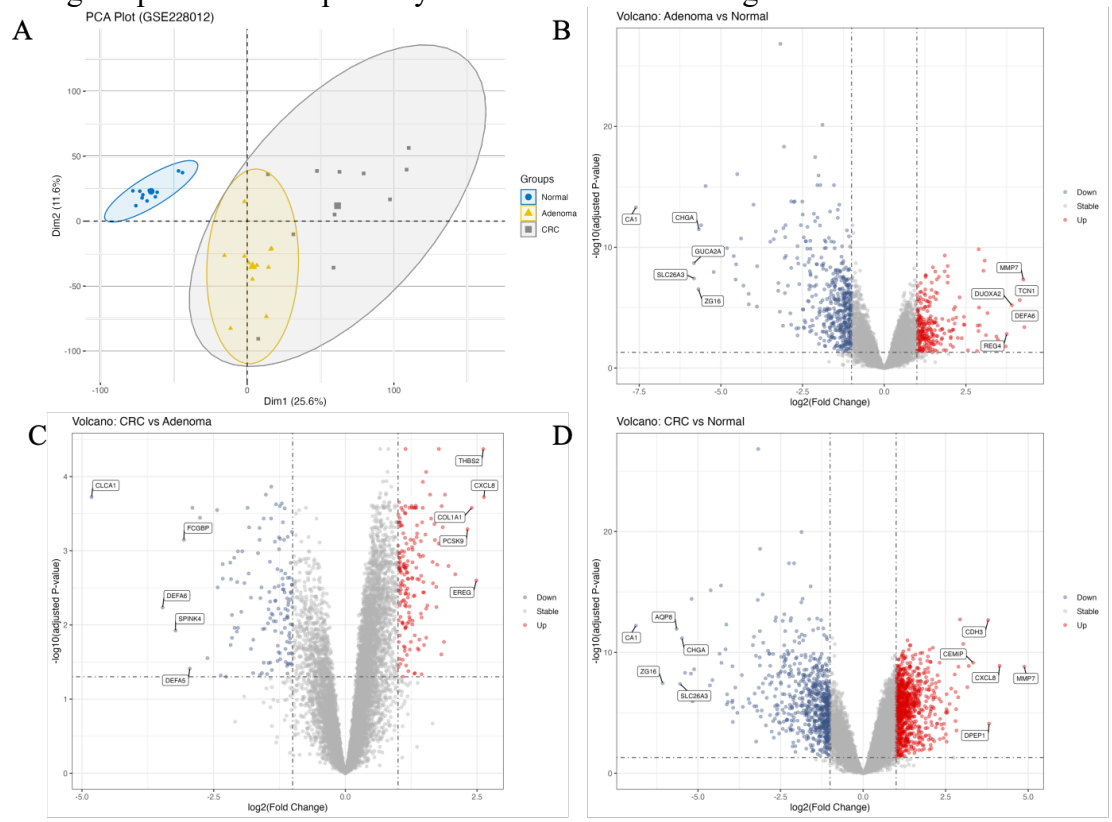


Figure 1. Global transcriptomic differences and differential gene expression across colorectal tumorigenesis.
(A) Principal component analysis (PCA) of 33 normal, adenoma, and stage-I CRC samples. Each point represents one sample; colors and shapes indicate tissue groups, and ellipses show group distributions. Axis percentages indicate variance explained by PC1 and PC2. (B–D) Volcano plots for adenoma versus normal (B), CRC versus adenoma (C), and CRC versus normal (D). Red and blue points indicate significantly upregulated and downregulated genes, respectively, while gray points are non-significant. Dashed vertical lines indicate |log2 fold change| = 1; statistical significance was defined as Benjamini–Hochberg-adjusted $P < 0.05$.

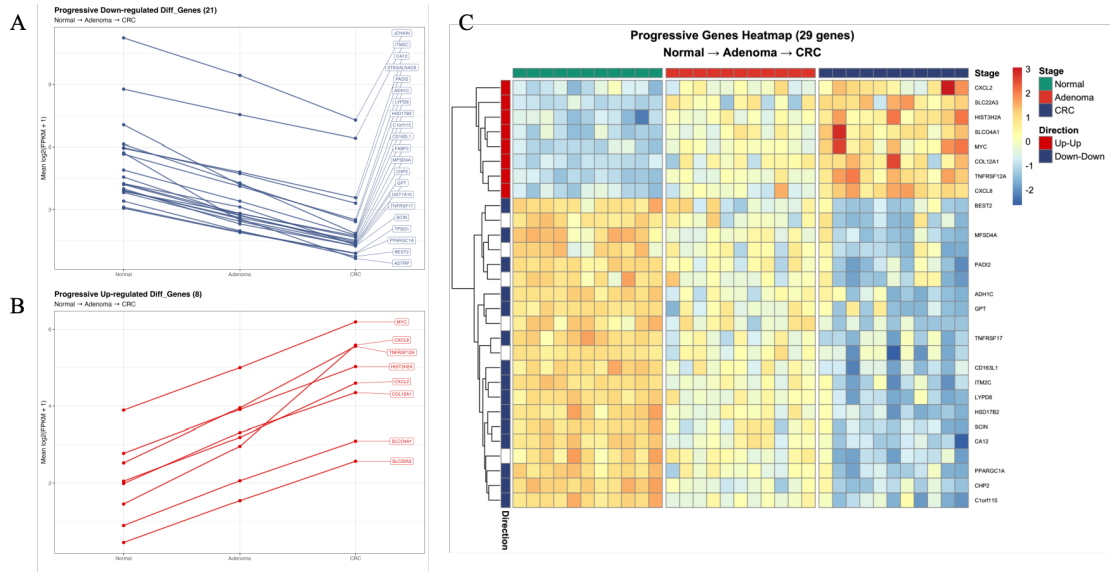


Figure 2. Progressive gene-expression changes across the normal–adenoma–CRC sequence.
(A) Heatmap of 29 genes showing consistent expression changes across the three pathological stages. Eight genes were progressively upregulated and 21 were progressively downregulated. (B) Mean-expression profiles of progressively downregulated genes. (C) Mean-expression profiles of progressively upregulated genes.

3.2 Functional changes associated with adenoma formation and carcinoma

The differential expression patterns identified across the normal–adenoma–CRC sequence suggested distinct biological changes at different stages of colorectal tumorigenesis. To investigate whether these transcriptional alterations were associated with biological mechanisms implicated in colorectal cancer development, we performed GO and KEGG enrichment analyses on the differentially expressed genes from each sequential comparison.

In the adenoma-versus-normal comparison, 239 of 4,540 tested GO biological processes, 21 of 381 cellular components and 68 of 766 molecular functions met the adjusted-P threshold (Table 1). Representative biological processes included humoral immune response, leukocyte chemotaxis, extracellular-matrix organization and detoxification, whereas collagen-containing extracellular matrix and apical plasma membrane were prominent cellular components; extracellular-matrix structural constituent and cytokine activity were representative molecular functions (Fig. 3A-C; Table 1).

KEGG analysis of the same contrast identified 10 significant pathways among 211 tested for upregulated genes and seven among 283 tested for downregulated genes (Fig. 3D-E; Table 1). Upregulated pathways included cytokine–cytokine receptor interaction, IL-17 signalling and NOD-like receptor signalling, while downregulated pathways included mineral absorption, bile secretion, fat digestion and absorption, and pentose and glucuronate interconversions (Table 1). Together, these results indicate that inflammatory recruitment and extracellular-matrix changes coexist with loss of differentiated epithelial and metabolic functions at the adenoma stage.

Table 1. Enrichment analysis of GO terms and KEGG pathways in the adenoma-versus-normal comparison.

Analysis category	Term	Gene count	Adjusted P value
GO biological processes	Humoral immune response	37	1.11E-07
GO biological processes	Leukocyte chemotaxis	32	1.11E-07
GO biological processes	Extracellular-matrix organization	38	1.94E-07
GO biological processes	Detoxification	25	4.84E-07
GO cellular components	Collagen-containing extracellular matrix	62	2.24E-19
GO cellular components	Apical plasma membrane	53	1.29E-14
GO molecular functions	Extracellular-matrix structural constituent	29	4.40E-10
GO molecular functions	Cytokine activity	27	8.18E-06
KEGG upregulated pathways	Cytokine–cytokine receptor interaction	16	6.09E-04
KEGG upregulated pathways	IL-17 signalling	9	6.09E-04
KEGG upregulated pathways	NOD-like receptor signalling	9	3.58E-02
KEGG downregulated pathways	Mineral absorption	16	1.15E-09
KEGG downregulated pathways	Bile secretion	11	2.22E-03
KEGG downregulated pathways	Fat digestion and absorption	8	2.22E-03
KEGG downregulated pathways	Pentose and glucuronate interconversions	6	1.81E-02

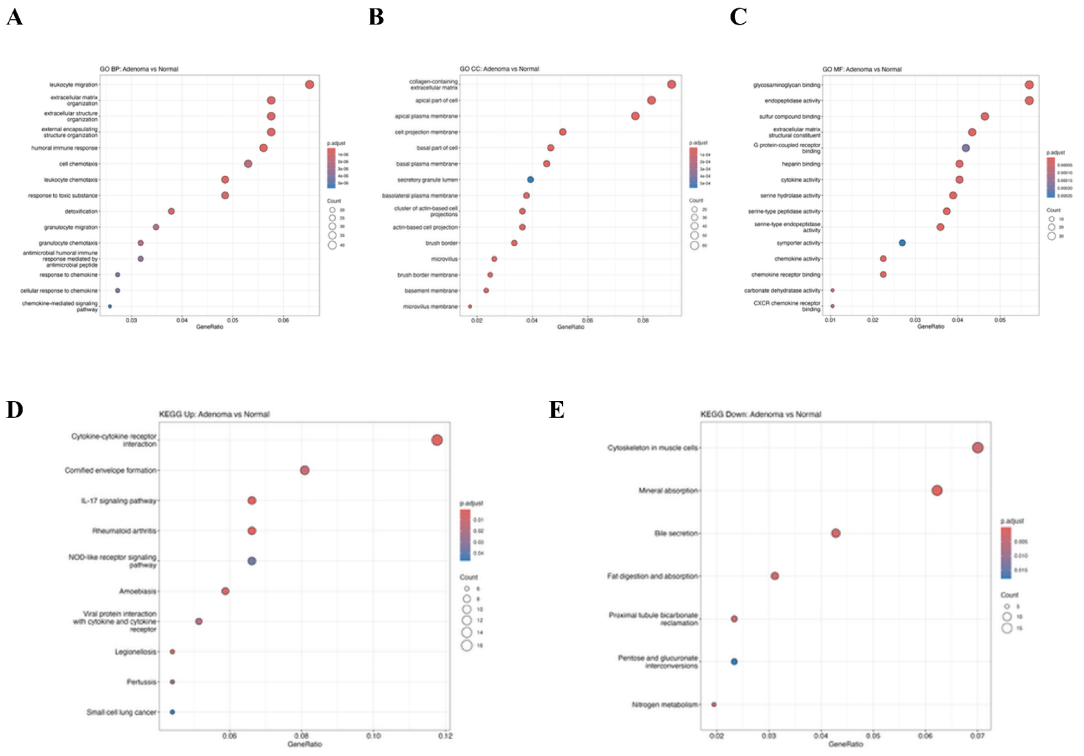


Figure 3. Functional enrichment for adenoma versus normal tissue.
(A–C) Representative GO biological-process, cellular-component and molecular-function terms. (D) KEGG pathways enriched among upregulated genes. (E) KEGG pathways enriched among downregulated genes. Dot size denotes gene count and color denotes the adjusted P value; panels show top-ranked terms, whereas all significant-term counts are reported in the text.

In the CRC-versus-adenoma comparison, 93 of 3,394 tested GO biological processes, 19 of 322 cellular components and 13 of 563 molecular functions met the adjusted-P threshold (Table 2). Representative biological processes included collagen fibril organization, steroid metabolic process, hormone metabolic process and extracellular-matrix organization. Collagen-containing extracellular matrix and collagen trimers were prominent cellular components, whereas extracellular-matrix structural constituent was the leading molecular function (Fig. 4A–C; Table 2). Genes contributing to these extracellular-matrix-related signals included COL1A1, COL3A1, COL5A1, COL5A2, COL11A1, COL12A1, FAP and POSTN.

KEGG analysis identified 14 significant pathways among 180 tested for upregulated genes and one among 168 tested for downregulated genes (Fig. 4D–E; Table 2). Upregulated pathways included protein digestion and absorption, proteoglycans in cancer, focal adhesion, integrin signalling, cell cycle, colorectal cancer and ECM–receptor interaction, whereas drug metabolism–cytochrome P450 was the only significant pathway enriched among downregulated genes (Table 2). Together, these results associate progression from adenoma to carcinoma with additional extracellular-matrix remodeling, cell–matrix adhesion and proliferative activity.

Table 2. Enrichment analysis of GO terms and KEGG pathways in the CRC-versus-adenoma comparison.

Analysis category	Term	Gene count	Adjusted P value
GO biological processes	Collagen fibril organization	11	3.05E-05
GO biological processes	Steroid metabolic process	19	4.05E-04
GO biological processes	Hormone metabolic process	16	4.05E-04
GO biological processes	Extracellular-matrix organization	19	4.05E-04
GO cellular components	Collagen-containing extracellular matrix	31	1.14E-10
GO cellular components	Collagen trimers	8	5.74E-08
GO molecular functions	Extracellular-matrix structural constituent	23	4.28E-13
KEGG upregulated pathways	Protein digestion and absorption	8	2.46E-03
KEGG upregulated pathways	Proteoglycans in cancer	10	2.56E-03
KEGG upregulated pathways	Focal adhesion	9	7.17E-03
KEGG upregulated pathways	Integrin signalling	8	7.17E-03
KEGG upregulated pathways	Cell cycle	8	7.17E-03
KEGG upregulated pathways	Colorectal cancer	5	3.89E-02
KEGG upregulated pathways	ECM–receptor interaction	5	3.94E-02
KEGG downregulated pathways	Drug metabolism–cytochrome P450	5	3.84E-02

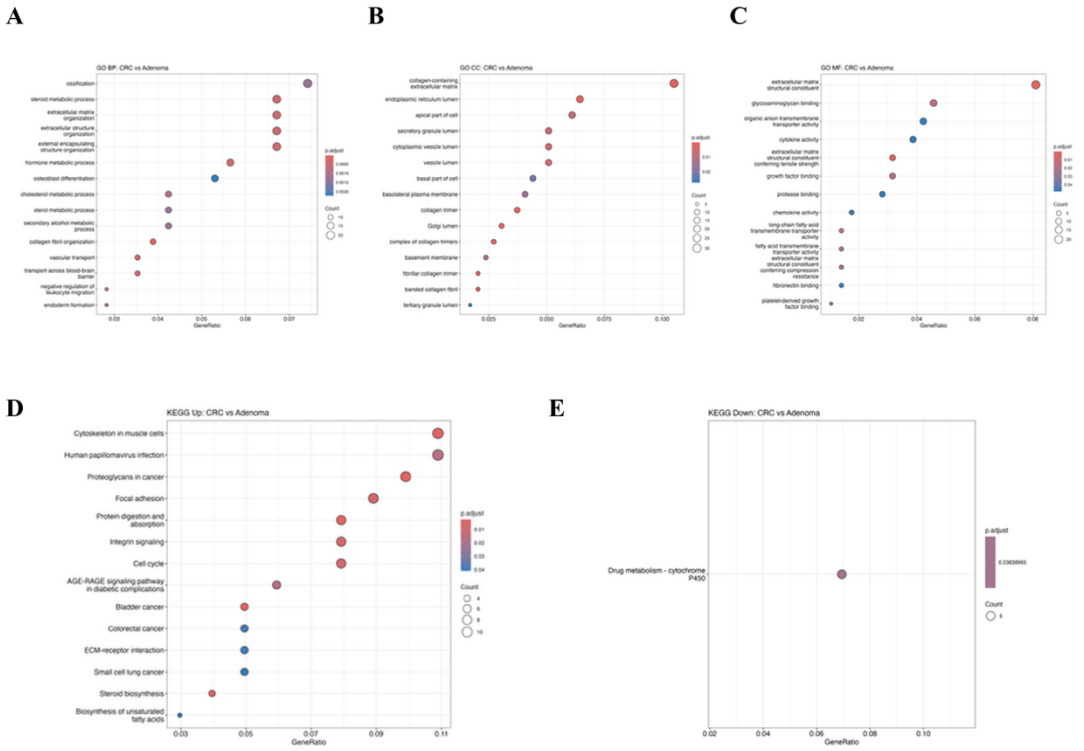


Figure 4. Functional enrichment for CRC versus adenoma.
(A–C) Representative GO biological-process, cellular-component and molecular-function terms. (D) The 14 significant KEGG pathways enriched among upregulated genes. (E) The single significant KEGG pathway enriched among downregulated genes. Dot size denotes gene count and color denotes the adjusted P value. GO panels display representative terms, whereas the KEGG panels display all pathways meeting adjusted $P < 0.05$.

3.3 PPI network of progression-associated genes

To determine whether the 29 progression associated genes identified from the sequential transcriptomic analysis represented an interconnected biological module, we examined the functional relationships among their encoded proteins using STRING. We further characterized the topology of this network to identify proteins occupying central positions within the progression associated module. This analysis was intended to provide a network-level view of the molecular alterations that persist across the normal–adenoma–CRC sequence and to complement the subsequent human protein interactome-based compound proximity analysis.

All 29 progression-associated proteins were represented in the STRING network, which contained 61 observed interactions compared with 28 expected interactions for a random protein set of the same size (Fig. 5A). The average node degree in our network was 4.21 and the local clustering coefficient was 0.577. The network showed significant PPI enrichment ($P = 2.99 \times 10^{-8}$), supporting functional connectivity among the progression-associated proteins.

We next examined network topology to identify proteins with potentially central positions within these 29 progression-associated protein modules. Degree centrality reflects the number of direct connections of a protein, whereas betweenness centrality reflects the extent to which a protein lies on shortest paths connecting other proteins. As shown in Fig. 5B, topological analysis identified MYC and CXCL8 as the most

highly connected nodes, each with a degree of 13. MYC showed the highest betweenness centrality (0.298), followed by CXCL8 (0.261), while both genes had a closeness centrality of 0.657.

The tendency of degree and betweenness centrality of key targets is highly correlated. However, some of the nodes possess high betweenness centrality values despite relatively low degree values. For example, CA12, with a degree of only 3, ranked ninth in betweenness centrality, underscoring its role as a bridge between otherwise separated network components. The top ten degree-ranked targets in the network are MYC, CXCL8, GPT, UGT1A10, PPARGC1A, TNFRSF17, ADH1C, CXCL2, HSD17B2, and FABP2, indicating their essential roles for the network.

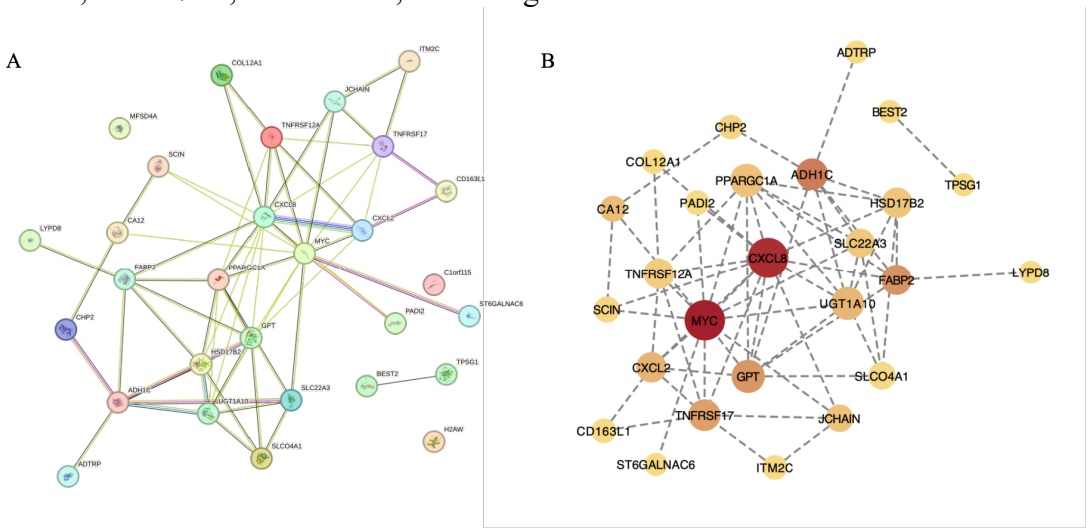


Figure 5. Protein–protein interaction network of progression-associated proteins. (A)The PPI network contained 29 nodes and 61 edges. Edges represent known or predicted functional associations among proteins. (B) Topological characteristics of the network nodes based on degree, betweenness centrality, and closeness centrality. MYC and CXCL8 showed the highest degree values.

3.4 Proximity between polyphenol targets and colorectal proteins

That the 29 progression-associated proteins form a connected network invites a practical question: which dietary polyphenols act on proteins close to this network? To address this, we used network proximity to rank polyphenols by how close their protein targets lie to the progression-associated proteins.

Polyphenols can be viewed as drugs in that they bind to specific proteins, affecting their ability to perform their normal functions. Previous work has shown that the network proximity between polyphenol targets and disease-associated proteins can predict known therapeutic effects of polyphenols, with closer proximity generally associated with a greater likelihood of affecting the disease phenotype (do Valle et al., 2021). Based on this framework, we examined whether the targets of individual polyphenols were unusually close to the 29 progression-associated proteins identified in our analysis, thereby prioritizing compounds potentially relevant to adenoma–carcinoma progression.

We therefore calculated the network proximity between polyphenol targets and proteins associated with 29 colorectal progression-associated genes to predict diseases for which specific polyphenols might have therapeutic effects, using the closest measure, dc, representing the average shortest path length between each polyphenol target and the nearest disease protein (Methods).

Among the 66 polyphenols compound analyzed, 12 met the predefined network-proximity threshold ($Z < -1.96$), indicating that their protein targets were located significantly closer to the progression-associated disease module than expected by chance. In this framework, greater network proximity suggests a stronger potential relationship between a compound and the molecular processes underlying adenoma–carcinoma progression. Resveratrol and trans-resveratrol shared the same PubChem CID (445154), identical target sets, and identical proximity scores, indicating that they represented the same compound. After merging these duplicate records, 11 unique compounds remained above the significance threshold.

In Fig. 6, we summarize the top 10 polyphenols that offer potential therapeutic effects according to network proximity calculations. The ten compounds with the most negative Z_{d_c} values were selected for subsequent structural characterization and molecular-docking analysis. Kaempferol showed the strongest network proximity (Z_{d_c}=-4.02; 37 mapped targets), followed by 4-methylcatechol (−3.67; 2 targets), quercetin (−3.61; 216 targets), punicalagin (−3.59; 1 target), umbelliferone (−3.49; 3 targets), esculetin (−3.12; 1 target), 2,3-dihydroxybenzoic acid (−2.56; 1 target), trans-resveratrol (−2.53; 63 targets), 1,4-naphthoquinone (−2.32; 4 targets) and phloretin (−2.16; 2 targets) (Fig. 6). Coumestrol also met the proximity threshold (Z_{d_c} =-2.07) but ranked eleventh among the compounds and was therefore not included in the downstream top ten set. (Fig. 6)

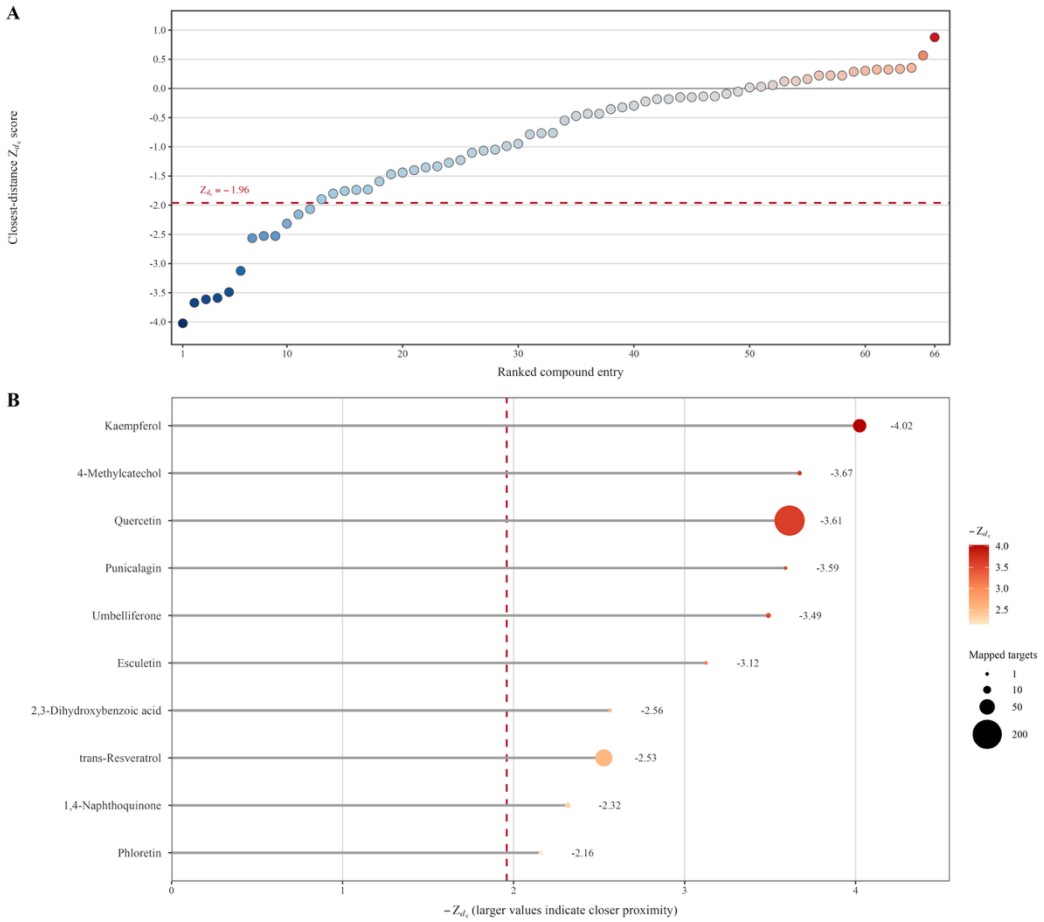


Figure 6. Network proximity between candidate compound targets and progression-associated proteins. (A) Distribution of network proximity $Z_{closest}$ scores across all 66 evaluated compounds. Dashed lines indicate the thresholds of $Z = -1.96$; more negative values represent closer-than-expected network proximity. (B) The ten compounds with the most negative proximity scores. Point size represents the number of mapped targets, and point color represents the magnitude of network proximity. More negative values indicate closer-than-expected network position.

Chemical identity and related evidence were listed within the same prioritization framework (Tables 3 and 4; Figs. 7 and 8) (Hastings et al., 2016; Kim et al., 2023). Compounds with limited CRC related evidence were retained as novel, underexplored candidates.

Table 3. Verified identities and structural classes of the curated top ten compounds

Compound	PubChem CID	Formula	MW (g/mol)	Class
Kaempferol	5280863	C15H10O6	286.24	Flavonol
4-Methylcatechol	9958	C7H8O2	124.14	Simple catechol
Quercetin	5280343	C15H10O7	302.23	Flavonol
Punicalagin	16129869	C48H28O30	1084.7	Ellagitannin

Umbelliferone	5281426	C9H6O3	162.14	Hydroxycoumarin
Esculetin	5281416	C9H6O4	178.14	Dihydroxycoumarin
2,3-Dihydroxybenzoic acid	19	C7H6O4	154.12	Phenolic acid/catechol
trans-Resveratrol	445154	C14H12O3	228.24	Stilbenoid polyphenol
1,4-Naphthoquinone	8530	C10H6O2	158.15	Quinone; not a polyphenol
Phloretin	4788	C15H14O5	274.27	Dihydrochalcone polyphenol

Table 4. Evidence-tier review of the curated top ten. Values in the third column are closest-distance Z score / number of targets mapped to the proximity interactome.

Rank	Compound	Z / targets	Past Article Review
1	Kaempferol	−4.02 / 37	Preclinical studies reported that kaempferol induced apoptosis and cell-cycle arrest in colorectal cancer cells. Additional studies suggested effects on glycolysis and chemotherapy resistance (Nejabati and Roshangar, 2022).
2	4-Methylcatechol	−3.67 / 2	Its sulfate correlated with tumour apoptosis markers after a black-raspberry intervention (Pan et al., 2015).
3	Quercetin	−3.61 / 216	Five FAP patients receiving curcumin plus low-dose quercetin had fewer and smaller polyps (Cruz-Correa et al., 2006).
4	Punicalagin	−3.59 / 1	Reduced viability and invasion in CRC cell lines (Sun et al., 2023).
5	Umbelliferone	−3.49 / 3	Reduced aberrant crypt foci and inflammatory markers in a DMH rat model (Muthu et al., 2016).
6	Esculetin	−3.12 / 1	Suppressed growth and metastasis in CRC cells and an orthotopic mouse model (Kim et al., 2018).
7	2,3-DHBA	−2.56 / 1	Inhibited CDK1/CDK6 in enzyme assays; proposed contribution similar to aspirin-associated CRC chemoprevention (Sankaranarayanan et al., 2020).
8	trans-Resveratrol	−2.53 / 63	In 20 presurgical CRC patients, parent compound/metabolites reached colorectal tissue and Ki-67 fell modestly (Patel et al., 2010).

Rank	Compound	Z / targets	Past Article Review
9	1,4-Naphthoquinone	−2.32 / 4	CRC evidence is old and indirect, linked to ex vivo/cell work on 1-naphthol and its quinone pathway (Wilson et al., 1985).
10	Phloretin	−2.16 / 2	Inhibited CRC-cell proliferation and β-catenin-associated outputs (Kapoor and Padwad, 2023).

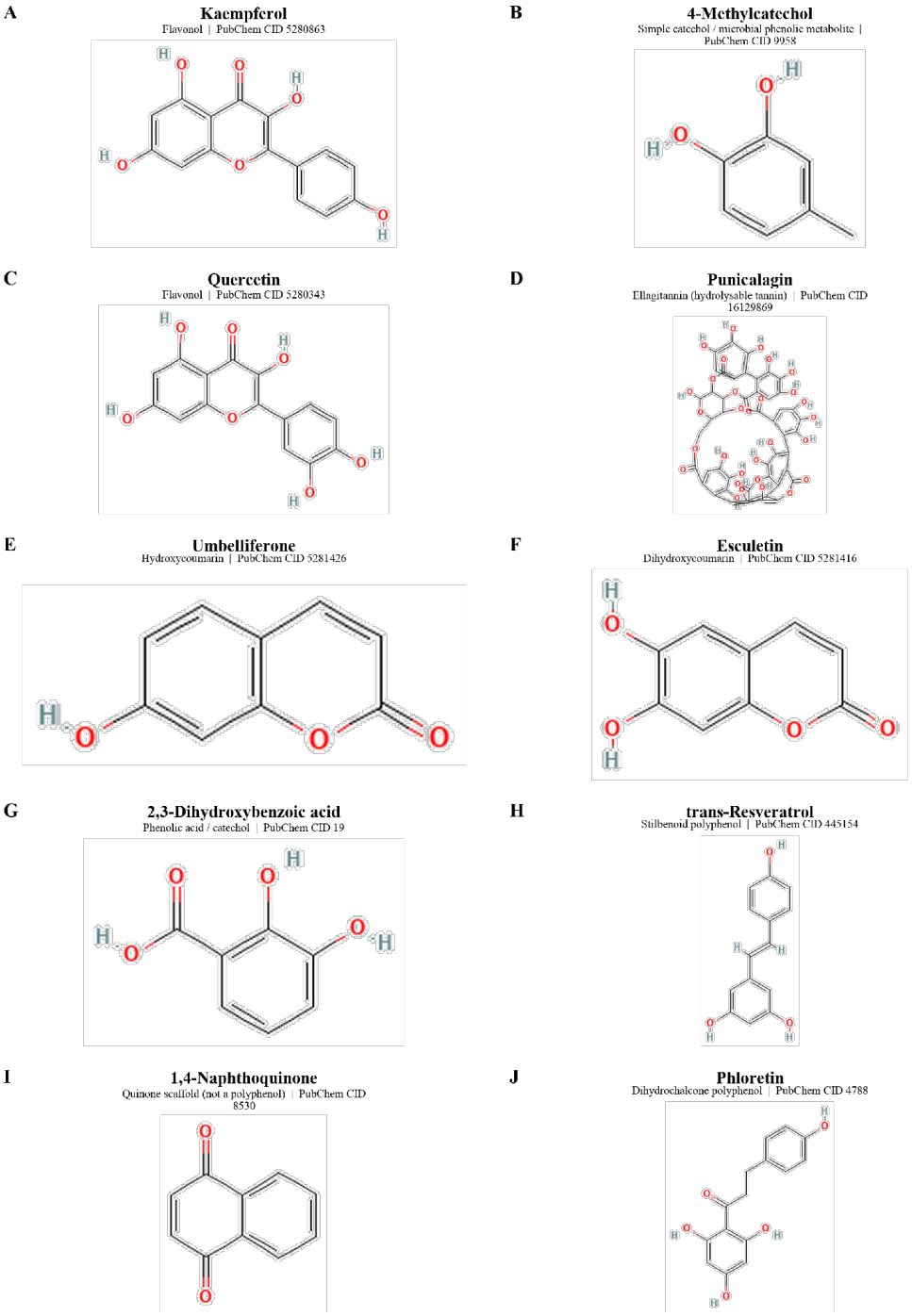


Figure 7. Verified two-dimensional structures and chemical classes of the curated top ten. Structures and identifiers were checked against PubChem and ChEBI records (Hastings et al., 2016; Kim et al., 2023). 1,4-Naphthoquinone is retained because it occurred in the screened dataset

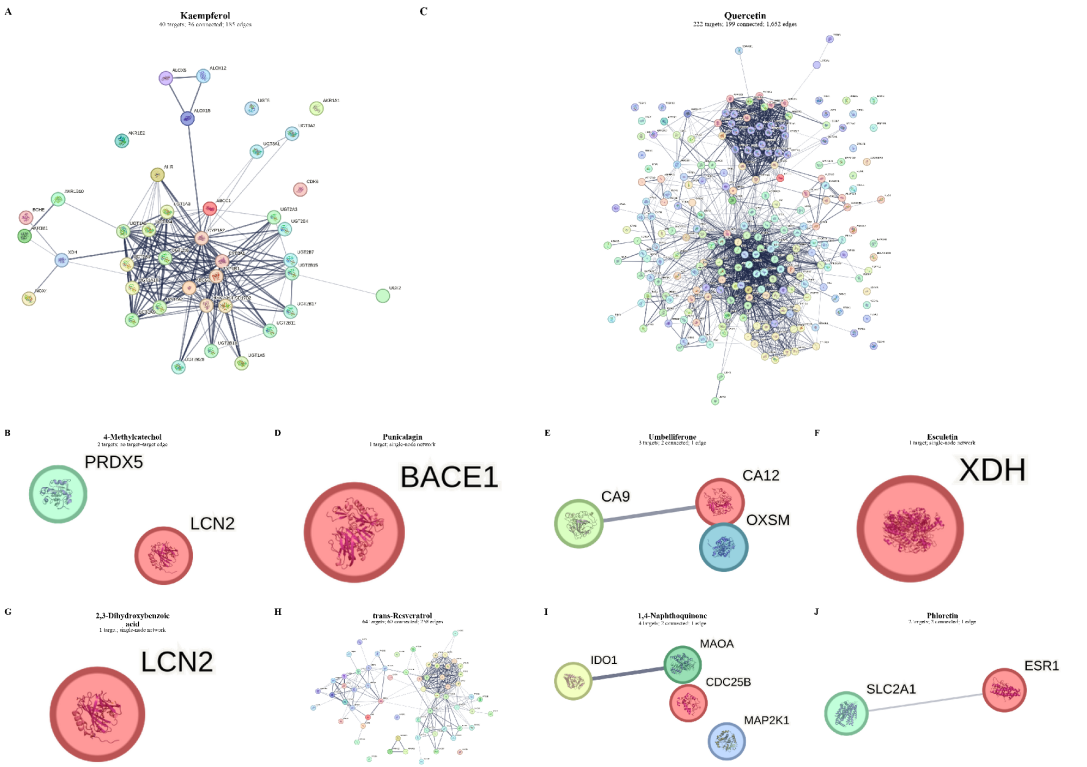


Figure 8. Functional associations among the recorded human protein targets of the ten prioritized compounds. Each panel represents one compound: (A) kaempferol, (B) 4-methylcatechol, (C) quercetin, (D) punicalagin, (E) umbelliferone, (F) esculetin, (G) 2,3-dihydroxybenzoic acid, (H) trans-resveratrol, (I) 1,4-naphthoquinone and (J) phloretin. Each node represents a human protein associated with the corresponding compound in the source *PolyphenolProteinInteractions* dataset used by do Valle et al. (2021), and node labels indicate human gene symbols. For each compound, all non-duplicated targets recorded in the source dataset were submitted to STRING v12.0 (Homo sapiens, taxon ID 9606). Edges represent direct or indirect functional protein–protein associations with a combined confidence score ≥0.400. This analysis describes functional connectivity within each compound-associated target set.

3.5 Molecular docking analysis

To complement the network-based prioritization with a structure assessment, molecular docking was performed for the ten prioritized compounds against selected progression-associated proteins. The analysis was used to examine whether these compounds could adopt plausible binding poses and to compare their predicted docking scores within each receptor system. CXCL8 and ADH1C were selected as representative targets for structural evaluation based on their relatively high connectivity within the progression-associated PPI network, with degree values of 13 and 7, respectively. Both also had experimentally resolved protein structures available for molecular docking.

For the complete 6WZM CXCL8-Fab assembly, the ten primary compound records produced best-mode scores ranging from -6.1 to -2.6 kcal/mol (5). Phloretin had the most favourable score within this receptor system (-6.1 kcal/mol), followed by trans-resveratrol (-5.9 kcal/mol), kaempferol (-5.7 kcal/mol), and quercetin and esculetin (both -5.6 kcal/mol). Within the predefined interface-centred search box, the representative phloretin pose was positioned between CXCL8 chain F and Fab heavy

chain C (Fig. 9A-B). The interaction map annotated a conventional hydrogen bond with Tyr105 of Fab chain C. Contacts involving CXCL8 chain F included a pi-donor hydrogen-bond interaction with Tyr13, pi-pi stacking with Phe21, and a pi-alkyl interaction with Leu43. An unfavorable donor-donor geometry involving Arg47 of CXCL8 chain F was also predicted.

For the ten primary compound records, docking against ADH1C produced best-mode Vina scores ranging from -8.3 to -5.5 kcal/mol (Table 5). Kaempferol had the most favorable score (-8.3 kcal/mol), followed closely by quercetin (-8.2 kcal/mol), phloretin (-8.1 kcal/mol), and trans-resveratrol (-7.7 kcal/mol). In the representative ADH1C-kaempferol pose, kaempferol was positioned in a pocket of chain A surrounded by Cys46, Arg47, Ser48, Thr178, Gly201, Gly202, Val203, Lys228, Val268, Ile269, Gln271, Val292, Gly293, Val294, Leu362, and Arg369 (Fig. 9C-D). Post-docking interaction visualization annotated a pi-cation contact with Arg47, a pi-sigma contact with Val203, and a carbon-hydrogen-bond contact with Gly201. Several additional residues contributed van der Waals contacts, whereas an unfavorable donor-donor geometry was annotated for Arg369. In summary, kaempferol, quercetin, phloretin and trans-resveratrol had the four most favorable scores against both CXCL8 and ADH1C. As Vina scores are only relative within each receptor, these results nominate kaempferol as the top candidate for experimental validation.

Table 5. AutoDock Vina scores for the ten selected compounds and the supplementary punicalagin structure record.

Compound	PubChem CID	ADH1C (kcal/mol)	CXCL8 (kcal/mol)
Kaempferol	5280863	-8.3	-5.7
4-Methylcatechol	9958	-5.5	-4.7
Quercetin	5280343	-8.2	-5.6
Punicalagin	16129869	-6.3	-2.6
Umbelliferone	5281426	-6.5	-5.5
Esculetin	5281416	-6.6	-5.6
2,3-Dihydroxybenzoic acid	19	-5.9	-4.7
Trans-resveratrol	445154	-7.7	-5.9
1,4-Naphthoquinone	8530	-7.0	-5.5
Phloretin	4788	-8.1	-6.1
Punicalagin	44584733	-6.2	+1.7

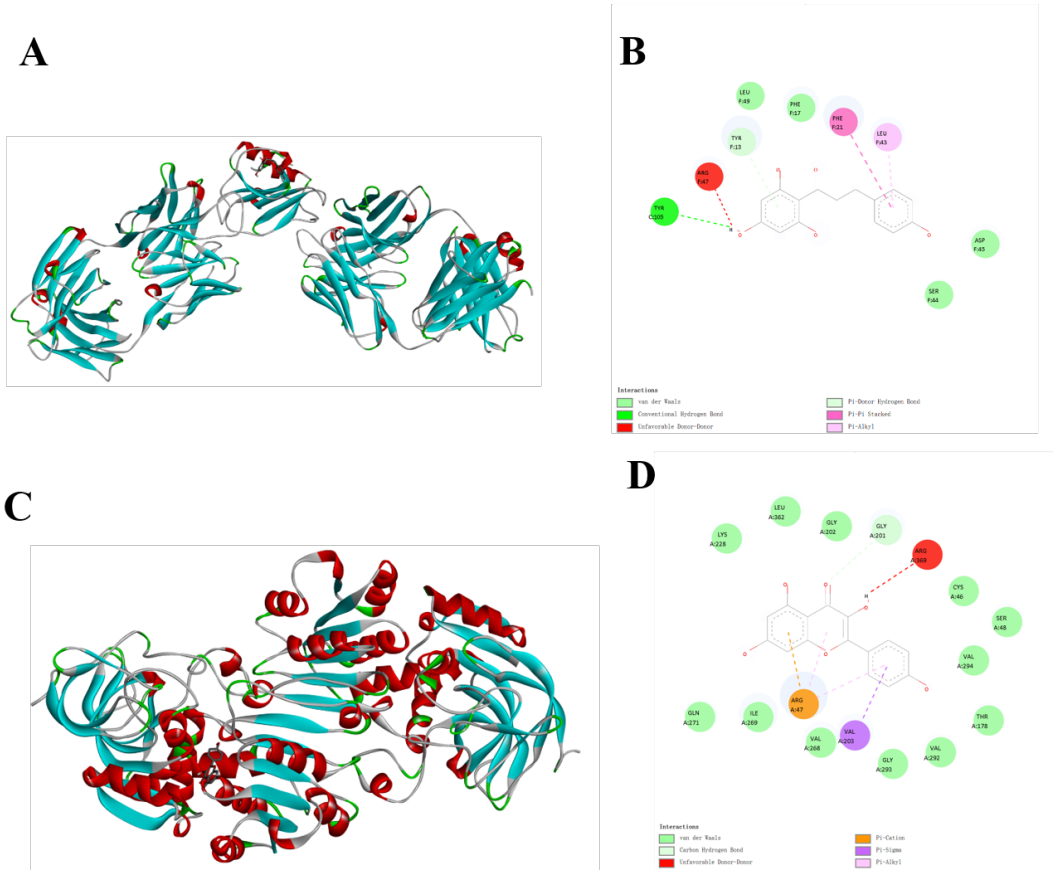


Figure 9. Representative docking interactions. (A) Overall view of phloretin docked within the predefined interface-centered search region of the complete 6WZM CXCL8-Fab assembly. (B) Two-dimensional interaction map of the 6WZM-phloretin pose. The best Vina score was -6.1 kcal/mol. (C) Overall view of kaempferol docked to ADH1C (PDB ID: 1HT0). (D) Two-dimensional interaction map of the ADH1C-kaempferol pose. The best Vina score was -8.3 kcal/mol. Interaction types shown in panels B and D are geometric annotations generated during post docking visualization.

4. Discussion

This study integrates cross-sectional transcriptomic comparisons across normal colorectal tissue, adenoma and CRC with interactome-based compound prioritization. The largest expression shift occurred between normal tissue and adenoma, whereas fewer genes differed between adenoma and CRC. Functional enrichment suggests that inflammatory recruitment and loss of mature epithelial transport and metabolic programs are established in adenomas, while carcinoma is associated with additional collagen organization, focal adhesion and cell-cycle activity.

The 29 progression-associated genes identified here are supported by previous research. MYC is a WNT-responsive oncogenic mediator whose deletion rescues major phenotypes caused by acute Apc loss in the mouse intestine (Sansom et al., 2007). CXCL8 has been associated with advanced stage, stromal abundance and metastatic features in human CRC tissue (Pennel et al., 2022). COL12A1 has been detected in desmoplastic stroma and at invasive fronts (Karagiannis et al., 2012). Conversely, progressive loss of CA12, HSD17B2, ADH1C, UGT1A10 and FABP2 is compatible with loss of differentiated epithelial functions.

Among the prioritized compounds, trans-Resveratrol had the strongest direct

human evidence, with both the parent compound and its metabolites detected in colorectal tissue and a modest reduction in tumour Ki-67 levels after eight days (Patel et al., 2010). Esculetin exhibited activity in an orthotopic CRC mouse model (Kim et al., 2018), and umbelliferone reduced aberrant crypt foci in a chemically induced rat model (Muthu et al., 2016). Kaempferol, punicalagin, 2,3-dihydroxybenzoic acid, and phloretin were supported primarily by cell-based or early preclinical studies (Yoshida et al., 2008; Sun et al., 2023; Sankaranarayanan et al., 2020; Kapoor and Padwad, 2023). Quercetin was evaluated only as part of a five-patient curcumin combination study, so its independent contribution could not be determined (Cruz-Correa et al., 2006). Caution is warranted for 4-Methylcatechol, which is a simple microbial phenolic metabolite; although its sulfate metabolite correlated with tumour apoptosis markers after a black-raspberry intervention, chronic high-dose dietary exposure caused forestomach and glandular-stomach tumours in rats (Asakawa et al., 1994; Pan et al., 2015). These compounds may remain in the computational output but should not be advanced as equivalent preventive candidates.

Based on molecular docking results, kaempferol was predicted to bind to the active site of ADH1C. Kaempferol exerts multifaceted chemopreventive and therapeutic effects against CRC. It modulates key epigenetic mechanisms, induces apoptosis, suppresses inflammation and oxidative stress, and inhibits tumor cell proliferation, thereby interfering with multiple hallmarks of CRC progression (Chen et al., 2026). In addition, kaempferol has been identified as an alcohol dehydrogenase (ADH) inhibitor that acts via mixed-type inhibition against ADH1C and other ADH isoforms (Su et al., 2025). By inhibiting ADH1C-mediated ethanol metabolism, kaempferol may reduce the formation of acetaldehyde, a carcinogenic metabolite, and thus potentially attenuate alcohol-associated colorectal carcinogenesis. Collectively, these findings suggest that kaempferol may prevent CRC through both direct tumor-suppressive mechanisms and indirect modulation of ethanol metabolism, although further experimental and clinical studies are warranted to validate these effects.

The principal limitations are the small sample size and lack of external transcriptomic or protein-level experimental validation. Before strong biomarker claims, the CRC–normal contrast should be repeated with patient blocking and the 29-gene module tested in an independent adenoma–CRC cohort. Compound validation should compare normal colon organoids, adenoma organoids and CRC models at exposure-relevant concentrations, followed by target-engagement, pharmacokinetic and toxicological assessment.

5. Conclusion

In this study, we applied an integrative network-medicine framework combining paired transcriptomic profiling with interactome-based compound prioritization to normal colorectal tissue, adenoma and CRC. Adenoma formation was characterized by inflammatory recruitment and loss of differentiated epithelial and metabolic programs, whereas progression to carcinoma was associated with additional extracellular matrix organization, focal adhesion, and cell-cycle activity. Among the 29 concordantly changed genes across sequential comparisons, network proximity analysis identified 11 unique candidate compounds, with kaempferol showing the strongest proximity. Molecular docking further predicted that kaempferol binds to the active site of ADH1C, suggesting a potential dual mechanism involving direct tumor-suppressive effects and inhibition of ADH1C-mediated ethanol metabolism. Collectively, this work provides a

novel network-based strategy for CRC prevention and highlights kaempferol as a promising candidate that merits further studies.

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Acknowledgments

As this paper draws to a close, I would like to express my sincere gratitude to all the teachers, family members, and friends who provided guidance, assistance, and support throughout this research.

This study originated from an experience that was deeply personal to me. My grandmother was diagnosed with colon cancer in 2019 and underwent prolonged treatment in the years that followed. She ultimately passed away from the disease in 2025. This experience had a profound impact on me and led me to develop an early interest in colorectal cancer, which later became one of the main reasons I chose this research topic. From her attending physician, I learned that most sporadic colorectal cancers do not arise suddenly but instead develop through a multistep process from normal mucosa to invasive carcinoma. This led me to focus on identifying gene-expression changes underlying this progression.

As I explored the literature further, I became familiar with approaches in network pharmacology and network medicine. At the same time, having grown up in Guangdong, I had already developed an interest in traditional Chinese medicine and bioactive compounds derived from natural products. I therefore began with a relatively broad investigation of naturally derived compounds. As my literature review progressed, I gradually narrowed the scope of the study to polyphenolic compounds.

The entire research project lasted approximately three months and was carried out independently by me. During the early stage, I focused on defining the research direction, reviewing literature related to the adenoma–carcinoma progression of colorectal cancer, and identifying suitable publicly available datasets. I then used publicly available transcriptomic datasets containing samples from normal colorectal tissue, adenomas, and early-stage colorectal cancer, obtained from the **Gene Expression Omnibus (GEO)** database, to perform differential gene-expression analysis. This analysis ultimately identified 29 progression-associated genes, which were used as the primary disease-related protein set for the subsequent network analysis.

In the next stage of the study, these progression-associated proteins and the known protein targets of polyphenols were mapped onto the human protein–protein interaction network. Network distance was then used to evaluate the proximity between different candidate compounds and proteins associated with colorectal cancer progression. During data preparation and analysis, I encountered several issues, including duplicate compound names and substantial variation in the number of known targets associated with different compounds. To address these problems, I carefully verified compound identities and repeatedly checked the corresponding target-protein information.

First, I would like to express my sincere gratitude to Associate Professor Pan Chen from Nanjing University of Chinese Medicine, whose research focuses on network pharmacology. When I first developed an interest in investigating colorectal cancer progression through network pharmacology, a teacher at my school recommended that I contact Professor Chen, who had previously been their colleague. Given his expertise in this field, I proactively reached out to him to discuss my research ideas and seek academic guidance.

Throughout the project, Professor Chen provided valuable guidance on the theoretical framework and methodological approaches of network pharmacology, helping me better understand how transcriptomic analysis, protein protein interaction networks, and compound target relationships could be integrated into my research(theoretical framework). His advice helped me refine my research direction and strengthen the scientific foundation of the study, while I remained responsible for conducting the computational analyses and preparing the manuscript.His guidance was provided by online meetings on an unpaid academic basis and involved no financial relationship.

I would also like to express my sincere gratitude to my school-based supervisor, too, **Mr. Lu Haoyang**, from the AP International Division of Guangdong Country Garden School. Mr. Lu holds a master's degree from Columbia University and has an academic background in biostatistics and mathematics. In this study, he mainly provided guidance on data processing in R and the analysis of gene-expression data from cancer samples. His guidance was likewise provided internally through the school on an unpaid academic basis and involved no financial relationship.

Both teachers served primarily in advisory roles by providing methodological and scientific guidance. I conceived the research project and carried out the experimental design, data analyses, curation, literature research, figure and table preparation, and writing of the manuscript.

The research process also involved many challenges. During the transcriptomic analysis stage, I had to learn data-processing and statistical methods in R with which I had little experience. When I moved on to network analysis, I needed to understand new concepts such as the human interactome, shortest-path network distance, randomization, and Z-scores. During the later stages of writing, I repeatedly assessed whether the predictions generated by the network analysis were consistent with existing experimental and clinical evidence. I addressed these challenges mainly by reading original research papers, discussing methodological questions with my supervisors, and revising and checking my analyses.

I would also like to give special thanks to my family and friends for their companionship and support throughout these four months of research. Because this project was directly connected to my own family experience, reviewing clinical information about the progression of colorectal cancer sometimes had a strong emotional impact on me, and at one point I found it difficult to continue writing. The understanding and encouragement of my family and friends helped me return to the research and persevere until the project was completed. For me, this study was not only an academic exploration, but also a process of learning how to confront a personal experience and search for answers through research.

Regarding the use of artificial intelligence tools, **OpenAI ChatGPT and Codex** were used as supporting tools during both the research process and the writing of this paper. Codex was mainly used for assistance with code revision, brainstorming approaches, answering technical questions, and organizing English-language content. ChatGPT was mainly used for preliminary discussions of the research topic, literature searches, and adjustments to paragraph structure. Any code or literature-related information provided by AI tools was independently checked by me before I decided

whether to use it. AI tools did not generate the research data and did not replace my own work in research design, numerical analysis, or molecular docking. All content in this paper was reviewed and verified by me, and I take full academic responsibility for the work.